# Fixed-length self-avoiding walks in grid boxes are eventually polynomial: proofs of twenty conjectured formulas from the OEIS

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**Abstract**

For fixed $k$ and $d$, let $a_{k,d}(n)$ denote the number of $k$-step self-avoiding walks in the $d$-dimensional grid box $\{0,\ldots,n-1\}^d$, summed over all starting cells, where—following R. H. Hardin's convention in the On-Line Encyclopedia of Integer Sequences (OEIS)—a $k$-step walk is a sequence of $k$ distinct pairwise-adjacent cells and each direction is counted separately. We prove that $a_{k,d}(n)$ agrees with a polynomial in $n$ of degree $d$ for all $n \geq 2k-1$, with coefficients determined by finitely many walk enumerations in boxes of side at most $2k-1$. Combining this with exact computations that close the finitely many cases below the threshold, we prove the conjectured ("empirical") formulas recorded in twenty OEIS entries: A188148–A188155 ($d = 2$, $3 \leq k \leq 10$), A187164–A187170 ($d = 3$, $3 \leq k \leq 9$), and A188785–A188789 ($d = 4$, $2 \leq k \leq 6$), conjectured by R. H. Hardin in 2011 and open since. All computations are reproducible, and key numerical certificates are formally verified in the Lean 4 proof assistant.

## 1 Introduction

The On-Line Encyclopedia of Integer Sequences (OEIS) [1] contains several thousand entries whose formulas are marked *empirical* or *conjectured*: closed forms, recurrences, or generating functions found by fitting the published terms, awaiting proof. A substantial family of such entries is due to R. H. Hardin, who in the early 2010s contributed extensive tables counting combinatorial configurations in finite grids, together with empirically observed formulas. Proving Hardin's conjectures has recently become an active pastime: Dougherty-Bliss and Kauers [2] proved his conjectures on "Hardinian arrays" via the transfer-matrix method, the key structural result being that the counts are eventually polynomial in the grid dimension; and individual entries have been settled directly on the OEIS, e.g. the 2023 proof notes of Corneth in entry A186864 (king's tours) and of Howroyd in entry A186851 (knight's tours), the latter by a bounding-box argument in the same spirit as ours.

This paper settles one such family in a uniform way. For fixed $k \geq 1$ and $d \geq 1$ consider the box $\mathcal{B}_n^d = \{0, 1, \ldots, n-1\}^d \subseteq \mathbb{Z}^d$, with cells adjacent when they differ by $\pm 1$ in exactly one coordinate, and let

$$a_{k,d}(n) \;=\; \#\big\{\, k\text{-step self-avoiding walks in } \mathcal{B}_n^d \,\big\},$$

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summed over all starting cells. Here a *$k$-step walk* is a sequence of $k$ *distinct* pairwise-adjacent cells $c_1, \ldots, c_k$ (thus traversing $k-1$ edges), and a walk and its reversal are counted separately. This is Hardin's convention, which we confirmed against the example section of OEIS entry A188148; we adopt it so that our statements match the OEIS entries verbatim.

Hardin computed $a_{k,d}(n)$ for the twenty pairs $(k, d)$ listed in Table 1 and conjectured, in each case, that $a_{k,d}(n)$ is given by an explicit polynomial of degree $d$ for all $n$ beyond a small explicit threshold $n_0(k, d)$. These conjectures (and the equivalent conjectured generating functions and linear recurrences added by C. Barker in 2018) have remained open. We prove all of them.

**Theorem 1.1** (Main theorem, informal)**.** *For every fixed $k$ and $d$, the count $a_{k,d}(n)$ agrees for all $n \geq 2k-1$ with a polynomial in $n$ of degree $d$ whose coefficients are determined by finitely many walk enumerations in boxes of side at most $2k-1$. For the twenty pairs $(k, d)$ of Table 1, the resulting polynomials coincide with the conjectured OEIS formulas, and direct computation for the finitely many $n$ between the conjectured threshold and $2k-1$ completes the proof of each conjecture exactly as stated.*

The structural part (Section 3) is an elementary locality argument: a walk of $m = k-1$ edges started at a cell $c$ only sees the box within $L^\infty$-distance $m$ of $c$, so the number of walks from $c$ depends only on the *clipped profile* of $c$—its distance to each face of the box, capped at $m$. Grouping cells by profile immediately yields eventual polynomiality. We do not claim this observation is deep; its value lies in the uniform statement with an explicit threshold, which reduces each OEIS conjecture to a finite, machine-checkable computation, and in the resulting complete and independently verifiable proofs of twenty long-standing conjectured formulas. This mirrors the situation in [2], where eventual polynomiality (there, via transfer matrices) was likewise the engine turning empirical formulas into theorems.

**Verification and formalization.** All computations were performed by two independent methods — direct enumeration and the profile method — which agree on an extensive common test set, and the profile method reproduces every term published in the OEIS entries (22–43 terms per entry). The polynomial coefficients are recovered by exact rational interpolation. In addition, all twenty conjectures have been formalized and machine-checked end to end in the Lean 4 proof assistant (Remark 4.2). Code and Lean sources are available at `https://github.com/d26521082/saw-polynomials`.

## 2 Definitions and conventions

Fix $d \geq 1$. Cells are points of $\mathbb{Z}^d$; two cells are *adjacent* when they differ by $\pm 1$ in exactly one coordinate. For $n \geq 1$ let $\mathcal{B}_n^d = \{0, \ldots, n-1\}^d$.

**Definition 2.1** (Walks; Hardin's convention)**.** *A* $k$-step self-avoiding walk *in $\mathcal{B}_n^d$ is a tuple $(c_1, \ldots, c_k)$ of distinct cells of $\mathcal{B}_n^d$ with $c_i$ adjacent to $c_{i+1}$ for $1 \leq i < k$. It traverses $m := k-1$ edges. Reversals are counted as distinct walks. We write $a_{k,d}(n)$ for the number of such walks, and, for a cell $c$, $f_n(c)$ for the number of walks with $c_1 = c$, so that $a_{k,d}(n) = \sum_{c \in \mathcal{B}_n^d} f_n(c)$.*

The term "$k$-step" for a walk on $k$ cells follows the OEIS entries; the example section of A188148 ("3-step" walks occupying three cells) fixes the convention unambiguously. Note $a_{1,d}(n) = n^d$ and $a_{2,d}(n) = 2d\, n^{d-1}(n-1)$ (each direction of each edge).

## 3 The eventual polynomiality theorem

Throughout this section $k, d$ are fixed and $m = k - 1$.

**Lemma 3.1** (Locality)**.** *For $c \in \mathcal{B}_n^d$ define the* clipped profile

$$\pi(c) \;=\; \big(\min(c^{(i)}, m),\; \min(n-1-c^{(i)}, m)\big)_{i=1}^{d} \;\in\; \big(\{0,\dots,m\}^2\big)^d.$$

*Then $f_n(c)$ depends only on $\pi(c)$; that is, there is a function $w$ on profiles, independent of $n$, with $f_n(c) = w(\pi(c))$ for all $n$ and all $c \in \mathcal{B}_n^d$.*

*Proof.* A walk of $m$ edges from $c$ changes each coordinate by at most 1 per step, hence stays inside $R(c) = \prod_{i=1}^{d}[\, c^{(i)} - \ell_i,\; c^{(i)} + r_i\,]$ where $\ell_i = \min(c^{(i)}, m)$ and $r_i = \min(n-1-c^{(i)}, m)$: indeed a cell of $\mathcal{B}_n^d$ within $m$ steps of $c$ along axis $i$ lies between $\max(0, c^{(i)} - m) = c^{(i)} - \ell_i$ and $\min(n-1, c^{(i)} + m) = c^{(i)} + r_i$. Conversely every cell of $R(c)$ lies in $\mathcal{B}_n^d$. Thus the walks from $c$ in $\mathcal{B}_n^d$ are exactly the walks from $c$ in the box $R(c)$, and translating $R(c)$ by $-c + (\ell_1, \dots, \ell_d)$ gives a bijection with walks in $\prod_i \{0, \dots, \ell_i + r_i\}$ started at $(\ell_1, \dots, \ell_d)$. The latter count depends only on $(\ell_i, r_i)_i = \pi(c)$. □

Each value $w(p)$ is a finite, explicitly computable integer: enumerate the walks of $m$ edges in a box of side lengths $\ell_i + r_i + 1 \le 2m+1$. By reflection and permutation of axes, $w$ is invariant under swapping $\ell_i \leftrightarrow r_i$ and permuting the $d$ pairs, which we exploit in computations.

**Lemma 3.2** (Census)**.** *Let $n \ge 2m+1$. Along one axis, as $x$ ranges over $\{0, \dots, n-1\}$, the clipped pair $\big(\min(x, m), \min(n-1-x, m)\big)$ takes the value $(i, m)$ exactly once for each $0 \le i \le m-1$ (namely $x = i$), the value $(m, j)$ exactly once for each $0 \le j \le m-1$ (namely $x = n-1-j$), and the value $(m, m)$ exactly $n - 2m$ times; no other values occur.*

*Proof.* If $x < m$ then, since $n-1-x > n-1-m \ge m$, the pair is $(x, m)$. Symmetrically if $x > n-1-m$ the pair is $(m, n-1-x)$ with $n-1-x < m$. The remaining $x \in [m, n-1-m]$, of which there are $n - 2m$, give $(m, m)$. The three ranges are disjoint because $m \le n-1-m$. □

**Theorem 3.3** (Eventual polynomiality)**.** *For all $n \ge 2m+1 = 2k-1$,*

$$a_{k,d}(n) \;=\; \sum_{S \subseteq \{1,\dots,d\}} (n-2m)^{|S|} \sum_{\substack{p = (p_1,\dots,p_d) \\ p_i = (m,m) \text{ for } i \in S \\ p_i \in E \text{ for } i \notin S}} w(p),$$

*where $E = \{(i, m) : 0 \le i < m\} \cup \{(m, j) : 0 \le j < m\}$ is the set of boundary pairs. In particular $a_{k,d}(n)$ agrees on $n \ge 2k-1$ with a polynomial in $n$ of degree (at most) $d$, whose leading coefficient is the count $w\big((m,m)^d\big)$ of $k$-step walks from a central cell of the $(2m+1)^d$ box.*

*Proof.* By Lemma 3.1, $a_{k,d}(n) = \sum_p N_n(p)\, w(p)$ where $N_n(p)$ is the number of cells with profile $p$. Profiles factor across axes, so by Lemma 3.2 $N_n(p) = \prod_{i=1}^{d} \nu_n(p_i)$ with $\nu_n((m,m)) = n - 2m$, $\nu_n(e) = 1$ for $e \in E$, and $\nu_n = 0$ otherwise. Collecting terms by the set $S$ of axes carrying the interior pair $(m, m)$ gives the displayed formula, a polynomial in $n - 2m$ (hence in $n$) of degree at most $d$. The degree-$d$ coefficient is the $S = \{1, \dots, d\}$ term, $w((m,m)^d) \ge 1$ (at least the straight walk exists). □

**Corollary 3.4** (Generating functions and recurrences)**.** *Fix* $(k, d)$ *and suppose* $a_{k,d}(n) = P(n)$ *for all* $n > n_0$*, with* $P$ *a polynomial of degree* $d$*. Then* $\sum_{n\geq 1} a_{k,d}(n)x^n = Q(x) + R(x)/(1-x)^{d+1}$ *for polynomials* $Q, R$ *with* $\deg R \leq d$*, where* $Q$ *absorbs the terms* $n \leq n_0$*, and* $a_{k,d}(n) = \sum_{i=1}^{d+1}(-1)^{i+1}\binom{d+1}{i}a_{k,d}(n-i)$ *for* $n > n_0 + d + 1$*. Hence the conjectured Barker generating functions and recurrences in the OEIS entries follow from the polynomial formulas.*

*Proof.* Standard: the $(d{+}1)$-st finite difference of a degree-$d$ polynomial vanishes. □

# 4 Proofs of the twenty conjectures

For each pair $(k, d)$ in Table 1 we proceed in three steps, all exact integer computations.

1. **Coefficients.** Compute $a_{k,d}(n)$ by the profile method (Lemmas 3.1–3.2: enumerate $w(p)$ for the finitely many profiles, then form the census sum) at $d+1$ points $n \geq 2k-1$, and recover the polynomial $P_{k,d}$ of Theorem 3.3 by exact rational interpolation. In all twenty cases $P_{k,d}$ *equals the conjectured OEIS polynomial, coefficient by coefficient.*

2. **Gap.** The OEIS conjectures assert validity for $n > n_0$ with $n_0 < 2k-1$. For the finitely many $n_0 < n < 2k-1$ we evaluate $a_{k,d}(n)$ exactly (the census sum is valid for every $n$, using the true multiset of clipped pairs for that $n$) and check $a_{k,d}(n) = P_{k,d}(n)$. All checks pass.

3. **Consistency.** Independently, direct enumeration over all starting cells agrees with the profile method wherever both were run, and the profile method reproduces *every* term published in the OEIS entry (22–43 terms per entry, fetched verbatim from the OEIS).

Steps 1–2, together with Theorem 3.3, constitute a complete proof of each conjectured formula; step 3 is an additional guard against implementation error. By Corollary 3.4 the conjectured generating functions and recurrences in the same entries follow.

**Remark 4.1** (Sharpness of the thresholds)**.** *The conjectured thresholds are exactly minimal: for every entry with* $n_0 \geq 1$ *we verified by exact computation that* $a_{k,d}(n_0) \neq P_{k,d}(n_0)$ *(e.g.* $a_{10,2}(3) = 0$ *while* $P_{10,2}(3) = -13704$*). For small* $n$ *the clipped neighborhoods of opposite faces overlap and the census of Lemma 3.2 no longer applies. That the true threshold* $n_0$ *is smaller than our guaranteed* $2k-1$ *reflects cancellation among boundary corrections, which the gap computation of step 2 captures; we did not seek a closed-form explanation.*

**Remark 4.2** (Formal verification)**.** *All twenty statements have been formalized and machine-checked end to end in Lean 4 with mathlib. For each* $d \in \{2, 3, 4\}$*, the locality lemma, translation invariance, the single-axis census, and the polynomial form of Theorem 3.3 are proved as theorems depending only on Lean's standard axioms (*`propext`*,* `Classical.choice`*,* `Quot.sound`*); the twenty OEIS entry theorems combine these with window constants and sub-threshold values evaluated by the compiler via* `native_decide`*, adding only the standard* `Lean.ofReduceBool` *trust assumption. For example, the Lean statement of A188148 reads: for all* $n \geq 2$*,* $(A\,2\,n : \mathbb{Z}) = 12n^2 - 24n + 8$*, where* $A\,m\,n$ *is the walk count defined from first principles. In addition, an independent list-based Lean implementation of the count is checked by* `native_decide` *against the published OEIS data.*

| OEIS | $(k,d)$ | $a_{k,d}(n)$ for $n > n_0$ | $n_0$ |
|---|---|---|---|
| A188148 | $(3,2)$ | $12n^2 - 24n + 8$ | 1 |
| A188149 | $(4,2)$ | $36n^2 - 100n + 56$ | 2 |
| A188150 | $(5,2)$ | $100n^2 - 360n + 272$ | 3 |
| A188151 | $(6,2)$ | $284n^2 - 1228n + 1152$ | 4 |
| A188152 | $(7,2)$ | $780n^2 - 3960n + 4432$ | 5 |
| A188153 | $(8,2)$ | $2172n^2 - 12500n + 16096$ | 6 |
| A188154 | $(9,2)$ | $5916n^2 - 38192n + 55600$ | 7 |
| A188155 | $(10,2)$ | $16268n^2 - 115548n + 186528$ | 8 |
| A187164 | $(3,3)$ | $30n^3 - 60n^2 + 24n$ | 1 |
| A187165 | $(4,3)$ | $150n^3 - 426n^2 + 312n - 48$ | 2 |
| A187166 | $(5,3)$ | $726n^3 - 2640n^2 + 2688n - 720$ | 3 |
| A187167 | $(6,3)$ | $3534n^3 - 15366n^2 + 19536n - 7056$ | 4 |
| A187168 | $(7,3)$ | $16926n^3 - 85380n^2 + 128832n - 57312$ | 5 |
| A187169 | $(8,3)$ | $81390n^3 - 463074n^2 + 801216n - 418032$ | 6 |
| A187170 | $(9,3)$ | $387966n^3 - 2452704n^2 + 4766544n - 2833872$ | 7 |
| A188785 | $(2,4)$ | $8n^4 - 8n^3$ | 0 |
| A188786 | $(3,4)$ | $56n^4 - 112n^3 + 48n^2$ | 1 |
| A188787 | $(4,4)$ | $392n^4 - 1128n^3 + 912n^2 - 192n$ | 2 |
| A188788 | $(5,4)$ | $2696n^4 - 9968n^3 + 11424n^2 - 4416n + 384$ | 3 |
| A188789 | $(6,4)$ | $18584n^4 - 82552n^3 + 119616n^2 - 64320n + 9984$ | 4 |

Table 1: The twenty proved formulas. In each case the polynomial is the $P_{k,d}$ of Theorem 3.3 and the threshold $n_0$ is exactly as conjectured in the OEIS entry.

# 5 Extensions

The argument uses nothing about the nearest-neighbor step set beyond finiteness, and nothing about the box being a cube.

1. *Other step sets.* For any finite step set $S \subseteq \mathbb{Z}^d \setminus \{0\}$ (king moves, knight moves, etc.) with $M = \max_{s\in S} \|s\|_\infty$, the same proof shows that the number of $k$-step $S$-walks (self-avoiding or not, or with any local constraint that is a function of the walk shape) in $\mathcal{B}_n^d$ is polynomial of degree $d$ in $n$ for $n \geq 2mM + 1$. This covers, for instance, Hardin's king-tour family A186861–A186870, one member of which (A186864) was proved by Corneth by a related argument.

2. *Rectangular boxes.* Replacing $\mathcal{B}_n^d$ by $\prod_i \{0, \ldots, n_i - 1\}$ yields, verbatim, a multivariate polynomial in $(n_1, \ldots, n_d)$ once every $n_i \geq 2m + 1$, proving the row-and-column formulas in the associated OEIS arrays (A188147, A187162, A188784).

3. *Walks with marked endpoints, closed walks.* Any statistic determined by the translation class of the walk and its clipped profile admits the same treatment; e.g. counts of self-avoiding *polygons* by perimeter in $\mathcal{B}_n^d$ are likewise eventually polynomial.

We intend the Lean formalization to cover the general statement in item 1.

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## References

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